Extending Stochastic Switching Models in a Probabilistic Bandit Task Using Stickiness-Based Decision Models

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Adaptive decision-making in uncertain environments requires integrating reward history to guide flexible action selection. Beron et al. (2022) showed that mice performing a probabilistic two-choice bandit task exhibit graded, history-dependent stochastic switching behaviour that is accurately captured by reinforcement learning and recursive logistic regression (RFLR) models incorporating choice-history bias ("stickiness"), whereas inference-based hidden Markov models (HMMs) exhibit excessive switching. Using the authors' publicly available analysis and modelling code, we first established a behavioural and modelling baseline for the original 80–20 reward condition. We then extended the analysis to a noisier 70–30 environment to examine how behavioural parameters adapt under increased reward uncertainty. Cross-validated parameter estimation revealed increased base stickiness and reduced surprise sensitivity under greater stochasticity, indicating environment-sensitive modulation of persistence. We further introduced nonlinear, prediction-error–dependent stickiness mechanisms and evaluated their effects using generative simulations and held-out predictive log-likelihood. Dynamic RFLR achieved the highest predictive performance among adaptive variants and improved reward-per-switch efficiency by reducing excessive switching while preserving reward acquisition. These findings suggest that stickiness functions not merely as a static bias, but as a volatility-sensitive control parameter that regulates the flexibility–stability trade-off under uncertainty.

INTRODUCTION

Adaptive decision-making in uncertain environments requires balancing behavioural flexibility with persistence. In probabilistic reversal learning tasks, agents must integrate reward history to infer changes in environmental contingencies while avoiding overreaction to stochastic outcomes. Such behaviour engages cortico-striatal circuits implicated in reinforcement learning and behavioural flexibility, including orbitofrontal cortex, medial prefrontal cortex, and dorsal striatum [2]. Dopaminergic signalling is thought to encode reward-prediction errors that guide value updating, while prefrontal regions contribute to monitoring environmental volatility and regulating behavioural persistence. Disruptions to these systems can alter switching behaviour, producing either excessive perseveration or oversensitivity to negative feedback. Thus, understanding how agents regulate persistence under stochastic reward conditions has both computational and biological significance. Beron et al. (2022) [1] addressed this problem in mice performing a dynamic two-choice bandit task. They showed that mouse behaviour deviates from that of an ideal Bayesian observer performing hidden Markov model (HMM) inference, particularly in its switching dynamics. Instead, behaviour was accurately captured by a recursive logistic regression (RFLR) model incorporating a choice-history bias ("stickiness") and a stochastic action policy. Importantly, these models achieved near-maximal reward rates despite deviating from theoretically optimal inference, suggesting that behavioural persistence functions as a stabilising mechanism under stochastic feedback. Although Beron et al. demonstrated that stickiness is necessary to explain mouse behaviour, the computational role of this parameter remains underspecified. In their formulation, stickiness was treated as a fixed bias that varies across reward contexts but does not adapt dynamically within a session. It therefore remains unclear whether stickiness reflects a static behavioural tendency or a control variable that is regulated online in response to environmental volatility. If stickiness buffers against stochastic reward fluctuations, it should scale with environmental uncertainty and respond to recent prediction errors. Testing this hypothesis requires extending the fixed-stickiness framework to include adaptive modulation mechanisms. In this study, we first establish a behavioural and modelling baseline using the framework introduced by Beron et al. (2022) under the original 80–20 reward condition. We then extend the analysis to a noisier 70–30 environment to examine how behavioural parameters scale with increased uncertainty. Finally, we introduce nonlinear, adaptive stickiness mechanisms and evaluate their effects on switching behaviour and reward efficiency using generative simulations and cross-validated predictive analyses. By integrating a validated baseline with targeted model extensions, we aim to clarify the computational role of stickiness as a dynamically regulated mechanism for behavioural persistence under uncertainty.

METHODS: BASELINE FRAMEWORK FROM BERON ET AL. (2022)

Behavioural signature of interest in the original study

Beron et al. (2022) demonstrated that mice performing a dynamic two-choice probabilistic bandit task exhibit graded, stochastic switching rather than deterministic win–stay/lose–switch behaviour. The key behavioural signature is the conditional switching probability $P(\text{switch} \mid \text{history})$, which increases with accumulating negative evidence but remains bounded below one. Despite stochastic feedback, mice achieve near-maximal reward rates, indicating efficient adaptation under uncertainty.

Baseline implementation and data sources

To establish a validated behavioural and modelling baseline, we used the publicly available dataset and analysis/model code released by Beron et al. (2022). We executed the authors' pipeline to regenerate the key behavioural summaries and model-comparison figures reported in the original study. All baseline figures were generated using the provided scripts/notebooks with fixed random seeds where applicable, and outputs were cross-checked against the published results for qualitative agreement (e.g., graded policy structure, model ranking, and HMM overswitching).

Computational models considered in Beron et al. (2022)

Beron et al. (2022) compared multiple model classes that differ in how reward history and environmental uncertainty are represented. We summarise these model families here to define the computational hypotheses evaluated in both the baseline validation and extension analyses.

Win–Stay/Lose–Switch (WSLS). A heuristic strategy that repeats rewarded actions and switches after unrewarded outcomes. WSLS predicts near-deterministic switching after losses and therefore fails to capture graded behavioural responses.

Reinforcement learning (RL). A model-free approach that incrementally updates action values via reward prediction errors, typically paired with stochastic action selection. RL captures gradual adaptation but does not explicitly represent latent states of the environment.

Inference-based models (HMM). A latent-state model that infers the current reward contingency from past outcomes using Bayesian belief updating. Inference models can adapt rapidly following reversals but may over-switch under stochastic feedback if stochastic reward omissions are over-interpreted as evidence for latent state change. Beliefs are updated according to:

$$P(s_t \mid o_{1:t}) \propto P(o_t \mid s_t) \sum_{s_{t-1}} P(s_t \mid s_{t-1}) P(s_{t-1} \mid o_{1:t-1}),$$

where $P(s_t \mid o_{1:t})$ is the posterior belief over latent reward states given observations $o_{1:t}$, and observations o_t correspond to trial outcomes (reward) conditioned on the chosen action at time t.

Recursive logistic regression (RFLR). A reduced model that predicts choices from accumulated choice–outcome history while incorporating a choice-history bias (stickiness). In the recursive formulation, the probability of choosing action a_t is:

$$P(a_t = 1) = \frac{1}{1 + \exp\left[-(\beta x_t + \alpha a_{t-1})\right]},$$

where x_t represents accumulated reward evidence, β is an inverse temperature controlling sensitivity to evidence, and α is the stickiness parameter biasing repetition of the previous choice. Additional analysis scripts were used to regenerate behavioural summary metrics reported in the original study, including conditional switching probabilities, event-triggered switching dynamics, and quantitative comparisons between model and mouse policies. These analyses confirmed qualitative agreement with the key behavioural signatures described by Beron et al. (2022) and established the modelling framework used for subsequent extensions.

METHODS: EXTENSIONS

Behaviour under increased uncertainty (70–30 condition)

Generative model simulations. Using cross-validated parameter estimates, we generated synthetic behaviour in both reward environments (80–20 and 70–30) under three model classes: RFLR, Ideal HMM (Thompson sampling), and Sticky HMM. To ensure identical reversal structure across models, simulations were run on the empirical latent state sequences extracted from the experimental sessions. Reward delivery probabilities were set to 0.8/0.2 (80–20) or 0.7/0.3 (70–30). Transition probabilities were set to the task transition parameter used in the generative simulations (p_{switch}), consistent with the probabilistic bandit structure. For each environment, RFLR parameters (α, β, τ) estimated via cross-validation were used to simulate behaviour. To assess parameter generalisation across environments, we additionally simulated the 70–30 condition using parameters fitted in the 80–20 environment.

Cross-validated parameter estimation across reward environments. To quantify how behavioural parameters depend on environmental uncertainty, the RFLR model was fit separately in the 80–20 and 70–30 reward environments. Parameter estimation was performed using 5-fold cross-validation at the session level to ensure robust generalisation and prevent information leakage between training and evaluation data. Sessions were randomly partitioned into five folds using a fixed random seed to ensure reproducibility. For each fold, model parameters (α, β, τ) were fit on four training folds by maximising the log-likelihood of the observed choice sequence under the RFLR model. Optimisation was implemented using stochastic gradient descent (10,000 iterations; learning rate = 0.01). The fitted parameters were then evaluated on the held-out fold to assess out-of-sample predictive performance. This procedure was repeated for all folds, and the final parameter estimate for each environment was taken as the mean of parameter estimates across folds.

Behavioural metrics and model comparison. Model behaviour was evaluated using summary statistics computed at the session level. For each simulated session, we computed the total reward obtained per session and the total number of switches per session. These metrics were used to characterise the reward-cognitive trade-off across models. To quantify switching dynamics around environmental reversals, we computed the probability of switching as a function of trial distance relative to block transitions. Switch probabilities were aligned to empirical state transitions and averaged across sessions to characterise cognitive lag and adaptation dynamics. Finally, parameter estimates (α, β, τ) were compared across the 80–20 and 70–30 environments to assess systematic shifts associated with increased reward uncertainty.

Nonlinear stickiness mechanisms

To extend the fixed-stickiness framework, we implemented dynamic modulation of the stickiness parameter (α) based on recent environmental feedback. These mechanisms were applied to the RFLR, HMM, and forgetting Q-learning (F-Q) architectures and evaluated using generative simulations.

Forgetting Q-learning (F-Q). The F-Q model maintains action values $Q_t(a)$ for each arm $a \in \{0, 1\}$, updated as

$$Q_{t+1}(a) = \begin{cases} Q_t(a) + k\big(r_t - Q_t(a)\big), & a = a_t, \\ (1-k)\, Q_t(a), & a \neq a_t, \end{cases} \tag{1}$$

where $k = 1 - e^{-1/\tau}$ is the learning rate (derived from the RFLR integration timescale τ) and the unchosen arm decays toward zero ("forgetting"). Actions are selected via softmax with stickiness:

$$P(a_t = 1) = \sigma\Big(\frac{Q_t(1) - Q_t(0)}{T} + \alpha\, c_{t-1}\Big),$$

where T is an inverse temperature and $c_{t-1} \in \{-1, +1\}$ encodes the previous choice.

Rolling window statistic Let W_t denote the set of indices corresponding to the most recent N trials up to time t (or fewer if $t < N$). We define the rolling statistic as

$$\bar{E}_t = \frac{1}{|W_t|} \sum_{j \in W_t} \mathrm{PE}_j. \tag{2}$$

This statistic was updated sequentially on each trial using a fixed-length rolling buffer.

Informational gating: signed prediction-error modulation (RFLR and HMM). For the RFLR and HMM models, stickiness was dynamically scaled as a function of recent signed prediction error (PE). On each trial $t > 0$, effective stickiness was computed via exponential gating:

$$\alpha_t = \alpha_{\text{base}} \exp\big(\gamma\, \bar{E}_t\big)\,, \tag{3}$$

where α_{base} is baseline stickiness, γ is surprise sensitivity, and $\bar{E}_t$ is the mean PE over an N-trial rolling window (defined above).

Prediction-error signals. Prediction-error signals were defined to match the implementation for each model class. For dynamic RFLR, we used an internal expected-reward proxy $\hat{r}_t = \sigma(\phi\, c_t)$ derived from the model's choice-weighted term c_t, and defined

$$\mathrm{PE}_t \;=\; r_t - \hat{r}_t, \qquad \mathrm{PE}_0 \;=\; r_0 - 0.5. \tag{4}$$

For Sticky HMM variants, PE_t was defined as the difference between the observed reward and the belief-weighted expected reward under the model's current posterior belief and emission probabilities. In the nonlinear Sticky HMM implementation, emission probabilities were parameterised asymmetrically with respect to chosen action, such that the reward likelihood under each latent state depended explicitly on the action taken at trial t. The belief-weighted expected reward was therefore computed using this action-conditioned emission structure, consistent with the computational implementation. For the F-Q nonlinear stickiness variant, the gating signal used the absolute value error $\varepsilon_t = |r_t - Q_t(a_t)|$, substituted for PE_k in the rolling window statistic defined above.

Belief reset under surprise (HMM variants). While dynamic modulation of α regulates motor-level persistence, standard Bayesian belief updating can be too rigid to capture rapid behavioural shifts during environmental reversals. To address this, we introduced belief-reset HMM variants inspired by Bayesian change-point detection [4, 5]. In these models, a surprise signal

$\omega_t \in [0,1]$ interpolates the standard Bayesian posterior toward a state of maximum uncertainty:

$$\pi_t = (1-\omega_t)\,\pi_t^{\text{raw}} + \omega_t \begin{bmatrix} 0.5 \\ 0.5 \end{bmatrix}, \tag{5}$$

where π_t^{raw} is the posterior obtained from standard Bayesian updating. The strength of the reset is governed by ω_t, which is derived from the rolling prediction-error statistic $\bar{E}_t$.

We tested two variants of this informational gating:

1. **HMM_reset (Asymmetric):** Operating under a disappointment-driven hypothesis, the surprise signal is triggered strictly by negative prediction errors (reward omissions):

$$\omega_t = 1 - \exp\big(\gamma\, \min(0,\, \bar{E}_t)\big). \tag{6}$$

   Only sustained omissions increase ω_t; unexpected rewards leave beliefs unchanged.

2. **HMM_resetv2 (Symmetric):** Aligning with a salience-driven hypothesis in which any large prediction error signals a potential structural change-point, the surprise signal is triggered by the absolute magnitude of the prediction error:

$$\omega_t = 1 - \exp\big(-\gamma\, |\bar{E}_t|\big). \tag{7}$$

   Both unexpected windfalls and unexpected omissions contribute equally to belief reset.

In both variants, dynamic stickiness modulation (α_t) and belief reset (ω_t) operate simultaneously: α_t regulates motor-level choice repetition via the exponential gating defined above, while ω_t independently modulates the cognitive belief state. This dual-track architecture allows the model to separately adjust behavioural persistence and environmental state estimation in response to surprise.

Multi-level gating framework and optimal evaluation

Model hypotheses. The adaptive extension evaluates distinct computational hypotheses for how agents use environmental "surprise" to regulate choice persistence in the 2ABT. We compare (i) a direct policy model where surprise gates motor persistence trial-by-trial (dynamic RFLR), (ii) a belief-based model where Bayesian state inference is combined with motor-level stickiness (Sticky HMM), and (iii) a temporal/value baseline where switching behaviour emerges from windowed value dynamics (FQ value-gated). In addition, belief-reset variants test whether omission-driven surprise triggers change-point detection and belief reinitialisation.

Cross-validated optimisation. To prevent overfitting, parameters were fit using 5-fold cross-validation over sessions. For each fold, model parameters were optimised on the training set and evaluated on held-out sessions using mean log-likelihood (Mean LL) as the primary performance metric.

Nested search over temporal parameters. Discrete temporal parameters (e.g., commitment window N) were selected via a nested search procedure. The outer loop enumerated candidate N values, and the inner loop performed continuous optimisation of model parameters (e.g., α_{base} and surprise sensitivity γ) using L-BFGS-B with bounds. For the value-gated F-Q baseline, γ was fixed at zero and only baseline stickiness was optimised. Parameter stability was assessed via across-fold variability, and the optimal N was selected via plurality vote across folds.

Simulation protocol and recorded outputs. Synthetic behavioural sessions were generated using the same probabilistic bandit task structure described above. Models were simulated across multiple sessions, and we recorded the dynamic stickiness trace, switching frequency, and (where applicable) value estimates (e.g., Q-values for F-Q). These outputs were used to assess how adaptive stickiness regulates the balance between persistence and flexibility under stochastic feedback.

Implementation details. All extension analyses were implemented in Python using custom code written for this project. Predictive evaluation used empirical mouse sessions grouped by reward condition, with each session sorted by trial order and partitioned into 5 shuffled cross-validation folds (fixed random seed 42). For each condition, the HMM hazard rate was estimated from empirical target-state block lengths, while β and τ were fixed to the cross-validated baseline means for that condition; for F-Q variants, the learning-rate parameter was set as $k = 1 - e^{-1/\tau}$. This yielded a controlled comparison of adaptive stickiness mechanisms while holding the baseline evidence-integration parameters fixed, rather than a fully unconstrained re-optimisation of every model component within each architecture. Generative comparisons were then run on the empirical target sequences extracted from the corresponding sessions, using the fitted adaptive model parameters together with the baseline parameters. We hypothesised that dynamic stickiness would improve switching efficiency by preventing excessive switching in response to stochastic reward fluctuations and, where evaluated against baseline models, improve alignment with mouse switching dynamics.

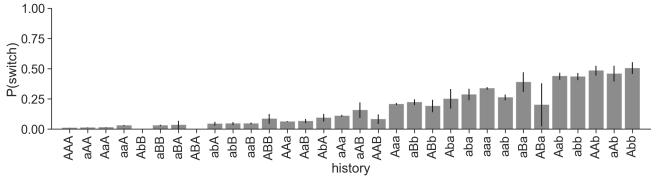


FIG. 1: **Mouse baseline exhibits graded, stochastic switching. Figure reproduced using the analysis pipeline from Beron et al. (2022).** Conditional switching probability $P(\text{switch} \mid \text{history})$ computed nonparametrically from mouse behaviour (history length = 3). Switching probability increases with accumulating negative evidence but remains bounded below one, demonstrating graded, stochastic rather than deterministic win–stay/lose–switch behaviour.

RESULTS

Validation of published behavioural signatures

We first verified that the publicly available analysis and modelling pipeline released by Beron et al. (2022) recovers the key behavioural and model-comparison results reported in the original study. Executing the authors' codebase regenerated the principal figures characterising graded, history-dependent stochastic switching in mice (Fig. 1) and the comparative performance of candidate models (Figs. 2–4). Conditional switching probabilities computed from mouse behaviour exhibited graded dependence on recent reward history and remained bounded below one (Fig. 1), confirming the stochastic switching structure described in the original study. Around contingency reversals, RFLR and logistic regression models captured gradual adaptation dynamics, whereas inference-based HMM variants exhibited sharper transitions and elevated switching (Fig. 2). Similarly, conditional switching structure and quantitative model–mouse agreement analyses (Figs. 3–4) confirmed that RFLR more closely matches the graded switching policy observed in mice, while HMM variants systematically overestimate switching probability across histories. Although this validation relied on the original implementation rather than an independent reimplementation, it establishes a consistent computational baseline and ensures that subsequent extensions are grounded in a verified modelling framework.

We next examined behaviour around contingency reversals. Mice exhibited gradual adaptation in both choice probability and switching rate. Logistic regression, RFLR, and forgetting Q-learning captured this gradual adjustment, whereas the HMM showed sharper transitions and elevated switching (Fig. 2).

The RFLR closely matched the graded switching policy observed in mice (Fig. 3), whereas the HMM systematically overestimated switching probability across many histories.

Scatter comparisons (Fig. 4) confirmed that RFLR provided the closest quantitative match to mouse switching behaviour, whereas HMM deviated substantially.

Together, these results are consistent with the main findings of Beron et al., demonstrating that mouse behaviour is best captured by models incorporating stochastic evidence integration and choice history bias.

Extension: Behaviour under increased uncertainty (70–30)

Having validated the original modelling framework, we next introduced a noisier 70–30 reward environment to examine how behavioural persistence adapts under increased stochasticity. Cross-validated parameter estimates revealed systematic modulation across environments (Fig. 5). Stickiness increased in the 70–30 condition, consistent with greater behavioural persistence under increased reward stochasticity.

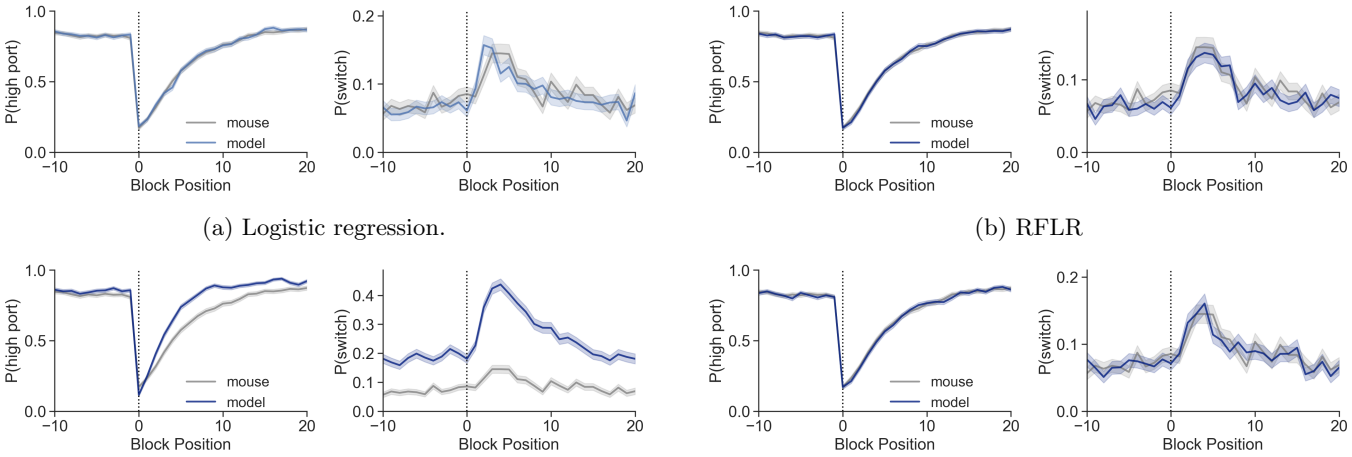
Behavioural efficiency revealed a clear reward–switch trade-off (Fig. 6). Although the HMM achieved slightly higher reward, it did so at the cost of substantially increased switching.

Reversal-aligned switching dynamics further demonstrated environment-specific effects (Fig. 7). Parameters fitted in the 70–30 condition captured gradual adaptation dynamics, whereas applying 80–20 parameters led to exaggerated switching.

Consistent with Fig. 6, the HMM occupied a high-switch, low-efficiency regime, whereas RFLR remained closer to mouse behaviour in both total reward and reward-per-switch efficiency.

Extension: Adaptive stickiness mechanisms

Building on the validated baseline, we implemented nonlinear, surprise-dependent modulation of stickiness to test whether dynamic regulation of persistence improves behavioural efficiency and predictive performance. We compared all adaptive variants jointly (Fig. 8). Relative to static RFLR, dynamic stickiness shifted behaviour toward lower switching rates while preserving reward, particularly under the noisier 70–30 condition. Across both reward environments, dynamic RFLR showed the clearest shift toward higher reward-per-switch efficiency relative to inference-only baselines. Reset-based Sticky HMM variants also reduced excessive switching relative to the inference-only HMM while preserving much of its reward performance. By contrast, the FQ-family extensions were more mixed and are best interpreted as sec-



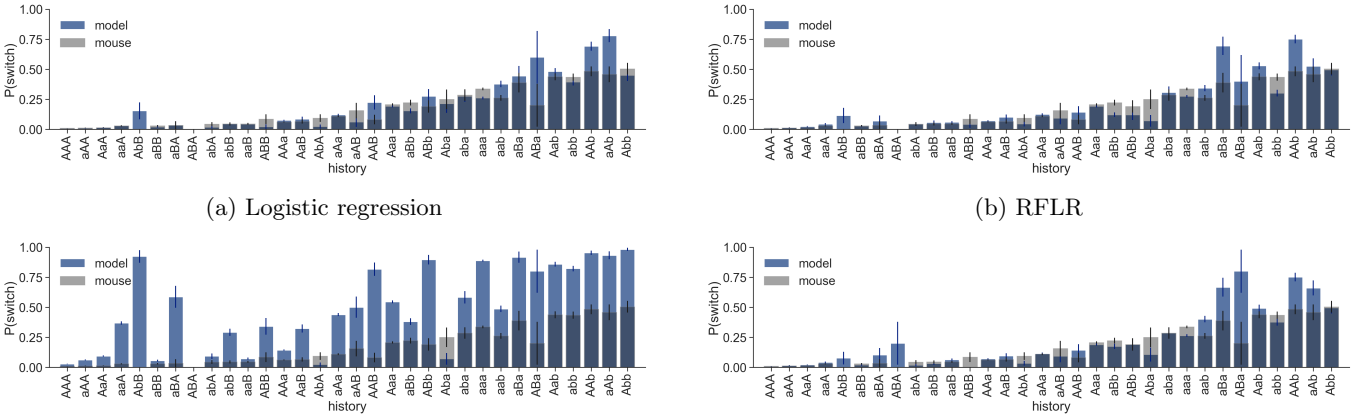
(a) Logistic regression.

(b) RFLR

(c) HMM

(d) Forgetting Q-learning

FIG. 2: **Model comparison around contingency reversals. Figures reproduced using the analysis pipeline from Beron et al. (2022).** Choice probability (left) and switching probability (right) aligned to block transitions. RFLR and logistic regression reproduce the gradual adaptation observed in mice, whereas the HMM exhibits sharper transitions and elevated switching, reflecting oversensitivity to stochastic reward omissions.



(a) Logistic regression

(b) RFLR

(c) HMM

(d) Forgetting Q-learning

FIG. 3: **Conditional switching structure reveals HMM overswitching. Figures reproduced using the analysis pipeline from Beron et al. (2022).** Mouse (grey) and model (blue) conditional switching probabilities across recent reward histories. RFLR closely matches the graded switching structure of mouse behaviour, while the HMM systematically overestimates switching across multiple histories.

ondary comparisons within the adaptive-model evaluation framework.

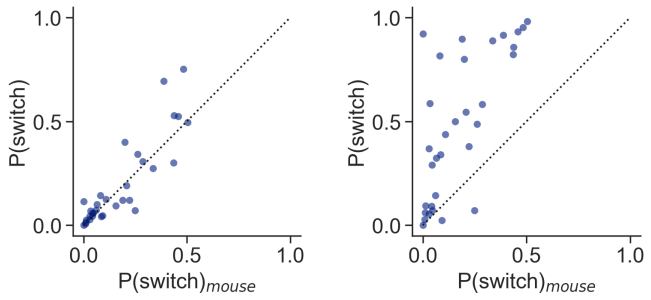
Finally, reversal-aligned generative transition dynamics revealed how adaptive mechanisms stabilise behaviour across environments (Fig. 9). Under the 80–20 condition, all models adapt rapidly following reversals. In contrast, under 70–30 uncertainty, adaptive stickiness produces more gradual, bounded switching dynamics, preventing oversensitive responses to stochastic losses.

Together, these generative results suggest that dynamic regulation of stickiness, especially in dynamic RFLR and to a lesser extent in reset-based Sticky HMM variants, can improve behavioural efficiency and produce switching dynamics that more closely resemble empirical mouse behaviour under increased reward stochasticity.

Predictive validation of adaptive stickiness mechanisms

We next evaluated whether adaptive stickiness mechanisms capture the fine-grained structure of mouse switching behaviour beyond reward-based efficiency metrics.

History-dependent switching structure. Figure 10(a,b) shows conditional switching probabilities across all 32 three-trial reward histories for the dynamic



(a) RFLR (b) HMM

FIG. 4: **Quantitative model–mouse agreement. Figures reproduced using the analysis pipeline from Beron et al. (2022).** Scatter plots comparing model-predicted and empirical $P(\text{switch} \mid \text{history})$ values. Points closer to the identity line indicate better agreement. RFLR demonstrates stronger quantitative alignment with mouse policy than the HMM.

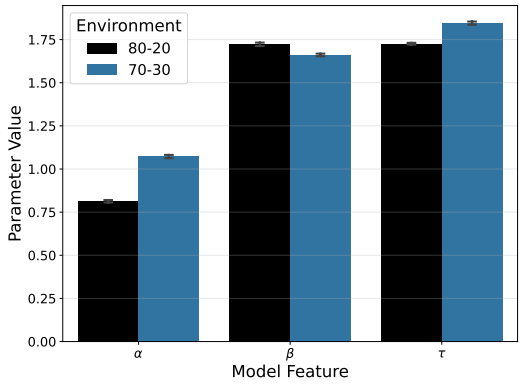


FIG. 5: **Stickiness increases under environmental uncertainty.** Cross-validated RFLR parameter estimates across 80–20 and 70–30 reward environments. Stickiness (α) increases systematically in the noisier 70–30 condition, indicating greater behavioural persistence under increased reward stochasticity.

RFLR model under both reward environments. Under 70–30 uncertainty, dynamic RFLR reproduces the graded structure of mouse switching probabilities, maintaining bounded switching even after streaks of losses. Under 80–20, switching remains sharply modulated by recent reward history, consistent with empirical behaviour.

Sticky HMM with symmetric reset exhibits similar graded structure (Fig. 10(c,d)), but produces slightly elevated switching following rare loss streaks, particularly under 70–30 uncertainty.

Quantitative policy agreement. We quantified model–mouse agreement by comparing predicted and empirical conditional switching probabilities across all reward–choice histories (Fig. 11). Dynamic RFLR exhibits strong alignment across both reward environments, whereas

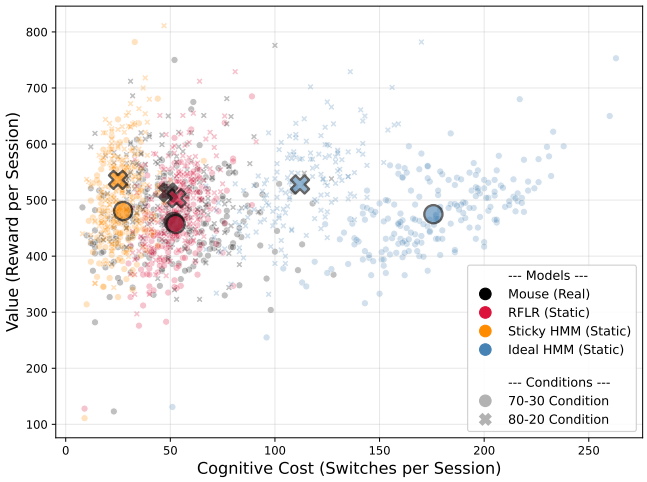


FIG. 6: **Reward–switch trade-off across environments.** Total reward per session plotted against switching frequency. While inference-based HMM achieves slightly higher reward, it does so at the cost of substantially increased switching, resulting in lower reward-per-switch efficiency compared to RFLR and mouse behaviour.

Sticky HMM with symmetric reset shows moderate but improved agreement relative to inference-only variants. These results indicate that the strongest adaptive variants, particularly dynamic RFLR, capture the detailed shape of mouse switching policy rather than merely reproducing aggregate reward statistics.

Reversal-aligned predictive dynamics. Finally, we evaluated predictive reversal dynamics across environments. As shown in Fig. 12, dynamic RFLR closely tracks mouse adaptation following contingency reversals under both 80–20 and 70–30 conditions, maintaining bounded switching and avoiding oversensitivity to stochastic outcomes. Sticky HMM with symmetric reset similarly stabilises reversal-aligned switching relative to inference-only models, though with slightly steeper transitions. The transient spike in predicted switching probability immediately following reversals reflects a known artifact of forced-choice (predictive) evaluation: the model detects early evidence of a contingency change before the mouse's motor response adjusts, producing a brief overshoot not present in generative simulations (Fig. 9) where the model controls its own actions.

Collectively, these predictive analyses indicate that dynamic RFLR, and to a lesser extent symmetric-reset Sticky HMM, reproduce the structured, graded, and environment-sensitive switching policy observed in mice while improving reward-per-switch efficiency relative to more reactive alternatives.

TABLE I: **Cross-validated predictive performance and fitted adaptive-stickiness parameters.** Number of free parameters k, mean held-out log-likelihood (Mean LL), fitted base stickiness α_{base} (mean ± SD across folds), surprise sensitivity γ (mean ± SD), and selected commitment window N for each model and reward environment. Higher Mean LL (less negative) indicates better predictive performance on held-out data. All adaptive variants except FQ_Value_Gated share $k = 3$, so the Mean LL ranking directly reflects predictive quality without confounding by model complexity.

Condition	Model	k	Mean LL	α_{base}	SD	γ	SD	N
80–20	RFLR	3	-0.1804	1.052	0.006	0.680	0.024	1
80–20	HMM_sticky	3	-0.2318	1.227	0.020	0.460	0.018	1
80–20	HMM_reset	3	-0.1984	0.606	0.005	0.883	0.014	1
80–20	HMM_resetv2	3	-0.1832	0.809	0.006	0.661	0.008	1
80–20	FQ	3	-0.2204	2.076	0.010	0.002	0.002	2
80–20	FQ_Value_Gated	2	-0.2204	2.077	0.010	0.000	0.000	2
70–30	RFLR	3	-0.1959	1.352	0.016	0.506	0.014	1
70–30	HMM_sticky	3	-0.2447	1.392	0.022	0.375	0.015	1
70–30	HMM_reset	3	-0.2094	0.818	0.008	0.854	0.009	1
70–30	HMM_resetv2	3	-0.2015	0.997	0.008	0.561	0.009	1
70–30	FQ	3	-0.2259	1.702	0.015	0.548	0.017	2
70–30	FQ_Value_Gated	2	-0.2278	2.172	0.007	0.000	0.000	2

Cross-validated predictive comparison of adaptive stickiness models

To complement reward–switch efficiency analyses, we compared adaptive model variants using held-out predictive log-likelihood from 5-fold cross-validation (Table I). Across both reward environments, dynamic RFLR achieved the highest predictive performance (least negative Mean LL; Table I). In the 80–20 condition, dynamic RFLR improved held-out log-likelihood by approximately 0.0028 per trial relative to the next-best adaptive variant (HMM resetv2), and by 0.0056 per trial in the 70–30 condition. Although modest at the single-trial level, these differences accumulate across sessions. Although adaptive variants introduce additional parameters (e.g., γ and commitment window size N), improvements in held-out likelihood indicate that gains reflect improved predictive structure rather than mere parameter flexibility. Parameter estimates also showed systematic context-dependence: in dynamic RFLR, base stickiness increased from $\alpha_{\text{base}} = 1.052 \pm 0.006$ in 80–20 to $\alpha_{\text{base}} = 1.352 \pm 0.016$ in 70–30, consistent with greater behavioural persistence under increased reward stochasticity. In contrast, surprise sensitivity decreased from $\gamma = 0.680 \pm 0.024$ (80–20) to $\gamma = 0.506 \pm 0.014$ (70–30), suggesting reduced trial-by-trial reactivity under higher uncertainty. Both shifts are robust across folds: the mean $\pm 2\,\text{SD}$ ranges for α_{base} are non-overlapping ([1.040, 1.064] vs. [1.320, 1.384]), as are those for γ ([0.632, 0.728] vs. [0.478, 0.534]). For the FQ family, the selected commitment window was $N = 2$, consistent with discretised value updating; predictive performance was tied in the 80–20 condition and slightly favoured the nonlinear FQ variant over the value-gated baseline in 70–30. Together, these results provide likelihood-based support for adaptive stickiness mechanisms and quantify how fitted persistence parameters shift with environmental uncertainty.

Normative interpretation: efficiency versus optimal inference

To contextualise adaptive stickiness mechanisms, we compared their behaviour to an inference-based benchmark (Ideal HMM with Thompson sampling), which approximates normative Bayesian state inference in the probabilistic bandit task. Although the Ideal HMM achieved marginally higher total reward (Fig. 6), it did so at the cost of substantially increased switching frequency, reflecting heightened sensitivity to stochastic reward omissions. In contrast, dynamic RFLR achieved comparable reward while maintaining lower switching rates and improved reward-per-switch efficiency. This pattern is consistent with the idea that adaptive stickiness may help regulate a trade-off between reward acquisition and switching cost, even though switching costs were not explicitly built into the fitted objective. In environments with stochastic feedback, excessive reactivity can degrade behavioural efficiency even if it remains inference-consistent. From this perspective, stickiness can be viewed as a candidate bounded-rational control mechanism: rather than implementing full latent-state inference, agents may regulate behavioural persistence to stabilise decisions under uncertainty. Adaptive modulation of stickiness allows persistence to scale with environmental volatility, increasing when feedback becomes unreliable and thereby preventing over-switching. Thus, dynamic stickiness may provide a compact algorithmic approximation to some of the efficiency trade-offs associ-

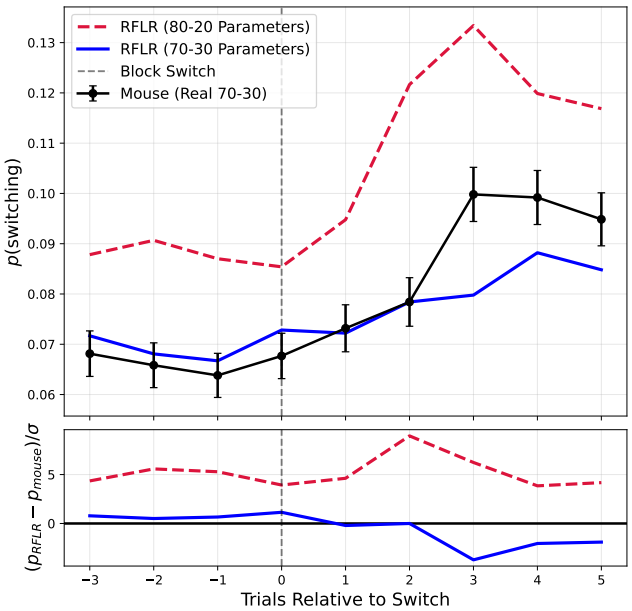


FIG. 7: **Cross-environment parameter transfer reveals the brittleness of static stickiness.** Upper panel: reversal-aligned switching probability $P(\text{switch})$ in the 70–30 condition. Circles with 1σ error bars show empirical mouse data; curves show generative RFLR simulations using parameters fitted in either the 80–20 (dashed) or 70–30 (solid) environment. The 80–20–parameterised model systematically overestimates switching and reacts prematurely to noise, whereas the 70–30–fitted model reproduces the empirical cognitive lag. Lower panel: standardised residuals $(p_{\text{RFLR}} - p_{\text{mouse}})/\sigma$, confirming that the 80–20 model diverges by multiple standard deviations from the empirical baseline.

ated with inference-based behaviour, sacrificing little reward while reducing switching cost. We treat this as an interpretive hypothesis supported by the present model comparisons rather than as a formal demonstration of optimality.

DISCUSSION

The central contribution of this work is not an independent reimplementation of the models introduced by Beron et al. (2022), but a computational validation and systematic extension of their published framework. By executing the publicly available analysis and modelling code provided by the authors, we confirmed that the reported behavioural signatures and model comparisons can be regenerated using the released pipeline. This validation step provided a reliable baseline upon which to build our extensions.

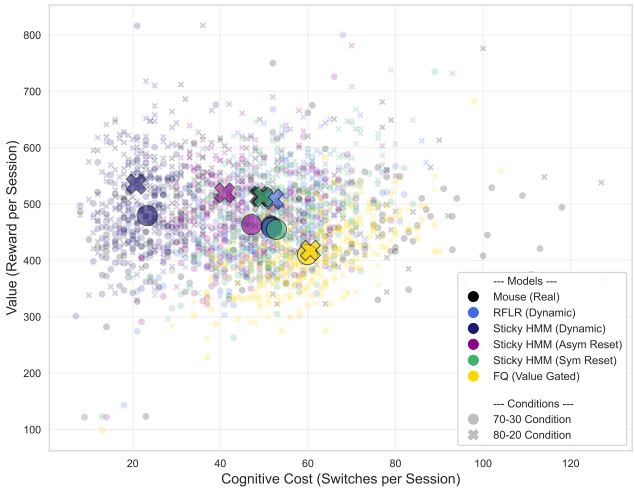


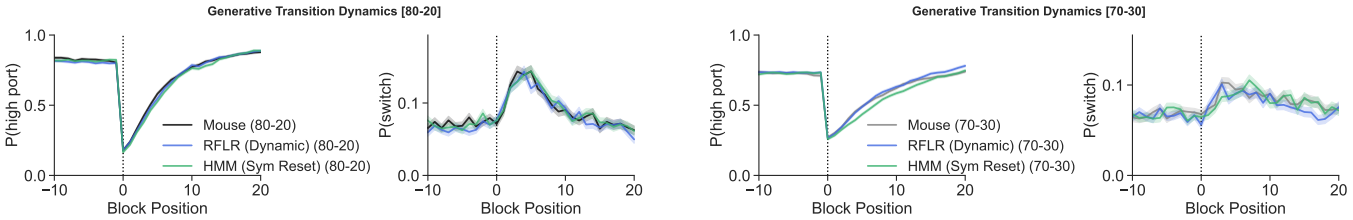
FIG. 8: **Adaptive variants can improve reward-per-switch efficiency.** Efficiency frontier comparison across adaptive variants. Dynamic RFLR shows the clearest improvement in reward-per-switch efficiency, while Sticky HMM reset models partially reduce the excessive switching seen in inference-only baselines.

Validation of stochastic switching structure

Using the released pipeline, we reproduced the graded, history-dependent switching behaviour originally reported in mice. Consistent with Beron et al. (2022), recursive logistic regression (RFLR) captured the bounded, stochastic switching structure more faithfully than inference-based HMM variants, which exhibited elevated switching under stochastic reward omissions. Although this validation relied on the original implementation rather than an independent derivation, it confirmed the internal consistency and robustness of the modelling framework.

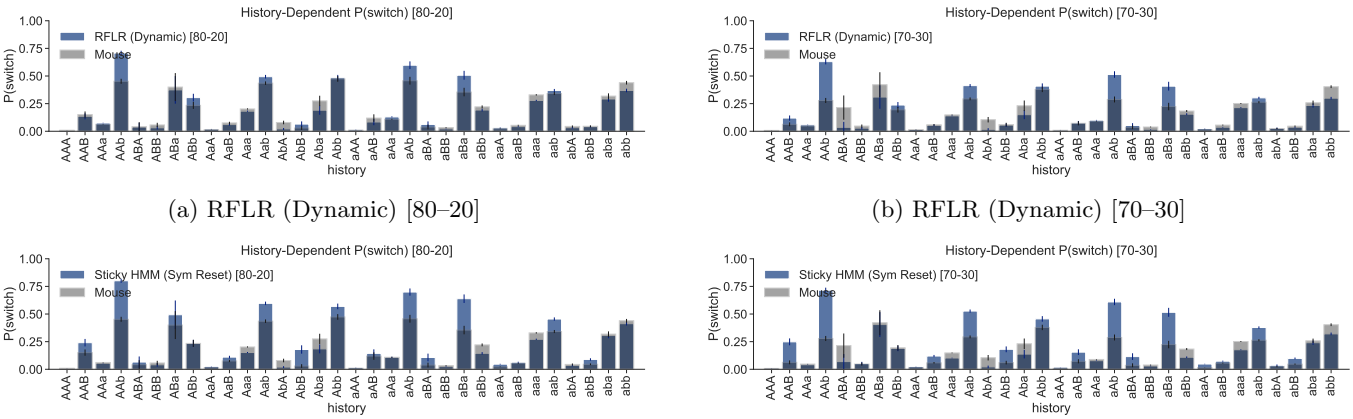
Stickiness increases with environmental uncertainty

Extending the task to a noisier 70–30 reward environment revealed systematic modulation of fitted parameters. Cross-validated estimates showed that base stickiness increased under greater reward stochasticity, while surprise sensitivity decreased, indicating reduced trial-by-trial reactivity in noisier conditions. Because this manipulation was not explored in the original study, these findings represent a novel extension of the framework. The observed parameter shifts suggest that behavioural persistence adapts to environmental volatility rather than reflecting a fixed bias. This interpretation is consistent with work showing that agents adjust learning and behavioural updating according to environmen-



(a) 80–20 (b) 70–30

FIG. 9: **Adaptive stickiness stabilises reversal-aligned dynamics.** Generative transition dynamics aligned to contingency reversals for both reward environments. Under increased uncertainty (70–30), adaptive stickiness produces more gradual and bounded switching, preventing oversensitive responses to stochastic losses.



(a) RFLR (Dynamic) [80–20] (b) RFLR (Dynamic) [70–30]

(c) Sticky HMM (Sym Reset) [80–20] (d) Sticky HMM (Sym Reset) [70–30]

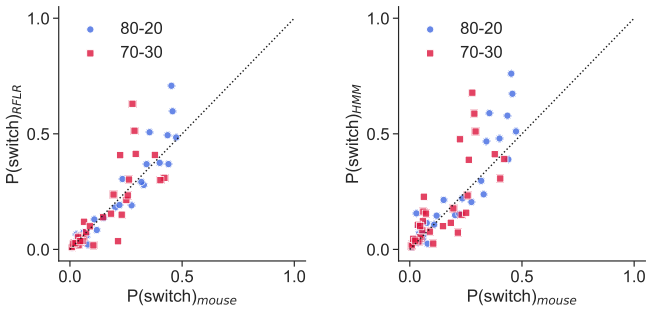
FIG. 10: **History-dependent switching structure for adaptive model variants.** Conditional switching probabilities across all 32 three-trial reward histories. Dynamic RFLR (a, b) preserves bounded, graded switching closely resembling empirical mouse policy structure. Sticky HMM with symmetric reset (c, d) reduces excessive switching relative to inference-only HMM, though switching remains slightly elevated following rare loss streaks, particularly under 70–30 uncertainty.

tal uncertainty and volatility, modulating the influence of recent outcomes when feedback becomes less informative [3, 5].

Adaptive stickiness improves efficiency and predictive performance

We further introduced nonlinear, surprise-dependent modulation of stickiness. Generative simulations demonstrated that adaptive regulation can reduce excessive switching while preserving reward acquisition, particularly under increased uncertainty. Importantly, cross-validated predictive evaluation showed that dynamic RFLR achieved the highest held-out log-likelihood among adaptive variants across both environments, with reset-based Sticky HMM variants showing weaker but directionally similar improvements over inference-only baselines. These results indicate that adaptive stickiness can improve not only reward-per-switch efficiency but also predictive fidelity to observed choice sequences.

Taken together, the comparative analyses reveal a consistent pattern across modelling frameworks, but the strength of evidence differs by model family. Inference-based models maximise reward but incur high switching cost due to heightened sensitivity to stochastic feedback. Static stickiness stabilises behaviour but lacks flexibility across environmental contexts. Dynamic RFLR provides the clearest evidence that adaptive modulation of persistence can recover much of the reward performance of inference-based models while maintaining bounded switching and improved efficiency; reset-based Sticky HMM variants move in the same direction, whereas the FQ-family extensions are more mixed. This positioning suggests that dynamic modulation of persistence may offer a computational middle ground between full Bayesian inference and simple heuristic repetition, capturing structured behaviour with reduced reactivity to noise, consistent with resource–rational accounts in which cognitive systems approximate optimal inference using computa-



(a) RFLR (Dynamic)

(b) Sticky HMM (Sym Reset)

FIG. 11: **Adaptive stickiness improves quantitative policy agreement.** Correlation between model-predicted and empirical conditional switching probabilities. Dynamic RFLR exhibits strong alignment across environments, while Sticky HMM reset variants show moderate but improved agreement relative to inference-only models.

tionally efficient strategies [6].

Stickiness as a computational control parameter

These findings support interpreting stickiness as a volatility-sensitive control parameter regulating the flexibility–stability trade-off in decision making, a balance closely related to the exploration-exploitation dilemma studied in reinforcement learning frameworks [7]. Rather than functioning as a static behavioural bias, stickiness appears to scale with environmental uncertainty, stabilising decisions when feedback becomes unreliable. From this perspective, adaptive stickiness may serve as a computationally efficient surrogate for some aspects of explicit latent-state inference, approximating belief-based behaviour while reducing sensitivity to stochastic noise.

Relation to neural mechanisms of flexibility and persistence

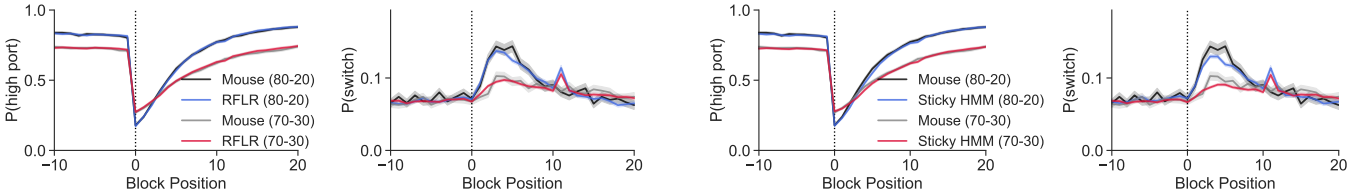
Although the present framework is algorithmic, the adaptive modulation of stickiness suggests possible links to neural mechanisms underlying behavioural flexibility. Probabilistic reversal learning is known to engage cortico-striatal circuits, including orbitofrontal cortex and dorsal striatum, which are implicated in updating action values and regulating behavioural persistence [2]. Dopaminergic signalling has been proposed to encode reward prediction errors [8], while prefrontal regions contribute to monitoring environmental volatility and controlling switching behaviour [3]. Importantly, the adaptive models introduced here operate on two functionally distinct control tracks. Dynamic RFLR modulates motor-level persistence (α_t), scaling the tendency to repeat recent actions according to prediction-error feedback. In contrast, the belief-reset HMM variants modulate the cognitive belief state (π_t), mixing the Bayesian posterior toward maximum uncertainty when sustained surprise indicates a potential environmental change-point. These tracks make different biological predictions: motor-level stickiness may correspond to gain modulation in striatal or motor circuits governing action repetition, whereas belief reset maps more naturally onto prefrontal volatility-monitoring mechanisms that detect structural changes in the environment. The observation that the symmetric-reset variant (HMM_resetv2) outperforms the asymmetric variant (HMM_reset) is consistent with salience-driven accounts in which the magnitude of prediction error, rather than its valence alone, drives change-point detection. From this perspective, dynamic stickiness may be viewed as a computational motif compatible with gain-control at the level of action selection, modulated by recent prediction error signals. Increased stickiness under higher uncertainty could reflect a neural strategy for dampening reactivity to stochastic negative outcomes, thereby preventing excessive switching. Conversely, reduced stickiness under stable contingencies would permit faster adaptation following true environmental changes. These links remain speculative: the present study does not identify neural mechanisms directly, but it does generate testable predictions for neurophysiological and perturbation studies.

Limitations

Several limitations should be acknowledged. First, model comparison was conducted using cross-validated log-likelihood and generative simulations, but was not embedded within a hierarchical framework across animals, limiting formal inference about individual variability. Second, switching costs were not explicitly parameterised within the objective function, meaning the observed efficiency trade-offs remain descriptive rather than formally optimised. Finally, the present framework operates at the algorithmic level and does not specify a mechanistic neural implementation.

Future directions

Future work could extend the present framework by embedding adaptive stickiness within a hierarchical modelling approach to quantify between-subject variability and environmental sensitivity more rigorously. Incorporating explicit switching or effort costs would allow a principled evaluation of efficiency trade-offs under re-



(a) Dynamic RFLR

(b) Sticky HMM (Sym Reset)

FIG. 12: **Predictive reversal dynamics confirm stabilising effect of adaptive stickiness.** Model-predicted choice and switching probabilities aligned to reversals across reward environments. Adaptive stickiness maintains bounded switching and avoids oversensitivity under increased uncertainty, closely tracking empirical mouse adaptation.

source constraints. More broadly, integrating adaptive stickiness mechanisms into biologically grounded circuit models and testing volatility-dependent modulation experimentally would strengthen links between computational theory and neural mechanisms of flexible decision-making.

CONCLUSION

In this project, we extended the computational framework developed by Beron et al. (2022) for modelling stochastic switching in probabilistic bandit tasks. Introducing a noisier 70–30 reward environment revealed systematic shifts in fitted persistence parameters under increased uncertainty. Adaptive stickiness mechanisms further improved reward-per-switch efficiency, with dynamic RFLR achieving the clearest and most consistent gains in held-out predictive performance across environments. These findings indicate that behavioural persistence is not merely a static bias, but a context-sensitive control variable that scales with environmental volatility. From a computational perspective, dynamic modulation of stickiness provides a parsimonious middle ground between full Bayesian inference and simple heuristic repetition, preserving reward performance while limiting excessive switching. Such regulation may approximate bounded-rational decision strategies that stabilise decision-making under stochastic feedback, although explicit cost-sensitive optimality remains for future work. These results also suggest possible neural relevance. Cortico-striatal circuits implicated in reversal learning and volatility monitoring could, in principle, support adaptive regulation of behavioural persistence through prediction-error–dependent gain control mechanisms. Framing stickiness as a volatility-sensitive control signal therefore offers a hypothesis linking algorithmic models of decision-making to candidate neural substrates of flexible behaviour. Overall, by extending the modelling framework to new environmental conditions and adaptive mechanisms, this work supports the view that volatility-sensitive regulation of persistence is a plausible computational strategy for balancing flexibility and stability in uncertain environments.

AUTHOR CONTRIBUTIONS

G.M., L.O., B.R., and N.Y. contributed equally to conceptualisation, analysis, and manuscript preparation.

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* All figures were generated using custom Python code written by the authors, unless otherwise stated (e.g., figures generated using the original analysis pipeline). The full analysis and figure-generation pipeline is available at `https://github.com/BOBRICE1/Stochastic-Switching-Bandits`.

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